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Course Code: MLA0401

Course Name: Deep Learning

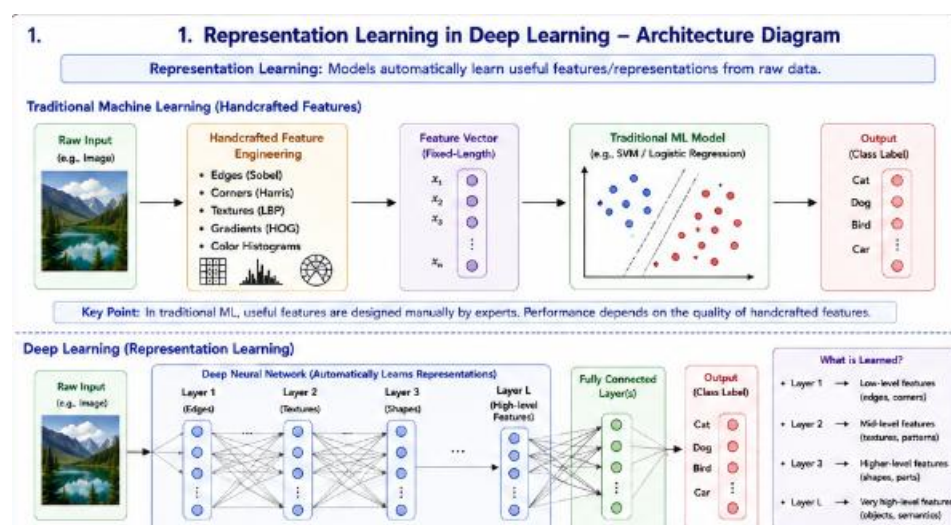
1. Representation Learning

Representation learning means a deep learning model automatically learns useful features from raw data instead of requiring humans to manually select features.

- In traditional ML, features are often manually selected.
- In deep learning, features are learned automatically through multiple layers.
- Early layers learn simple features; deeper layers learn complex features.
- Example: In an image, the model learns edges $\rightarrow$ shapes $\rightarrow$ objects.
- In NLP, it can learn words $\rightarrow$ phrases $\rightarrow$ sentence meaning.

Traditional ML: Raw data $\rightarrow$ Manual features $\rightarrow$ ML model $\rightarrow$ Output

Deep Learning: Raw data $\rightarrow$ Automatically learned features $\rightarrow$ Output



2. Width and Depth of Neural Networks

Width means the number of neurons in a layer.

- More width means more neurons can learn different patterns at the same level.
- It increases the model's learning capacity.
- Too much width increases parameters, memory usage, and overfitting risk.

Depth means the number of hidden layers.

- More depth allows the network to learn more complex hierarchical features.
- Early layers learn simple features.
- Later layers learn more abstract features.

Width

More neurons in a layer

Learns more patterns at the same level

Increases parameters

Can increase memory usage

Depth

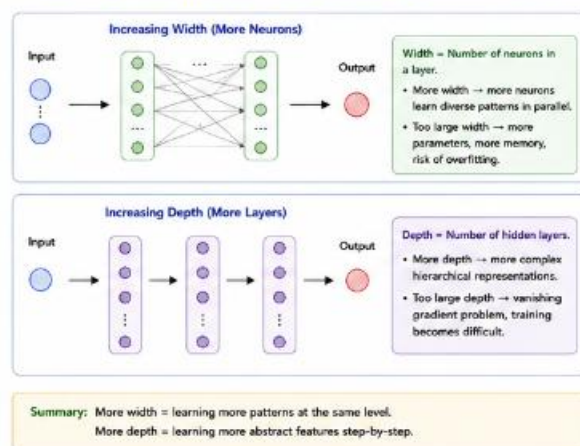
More hidden layers

Learns features step-by-step

Increases network complexity

Can cause training difficulties

2. Width and Depth of Neural Networks – Architecture Diagram



3. ReLU vs LReLU vs EReLU

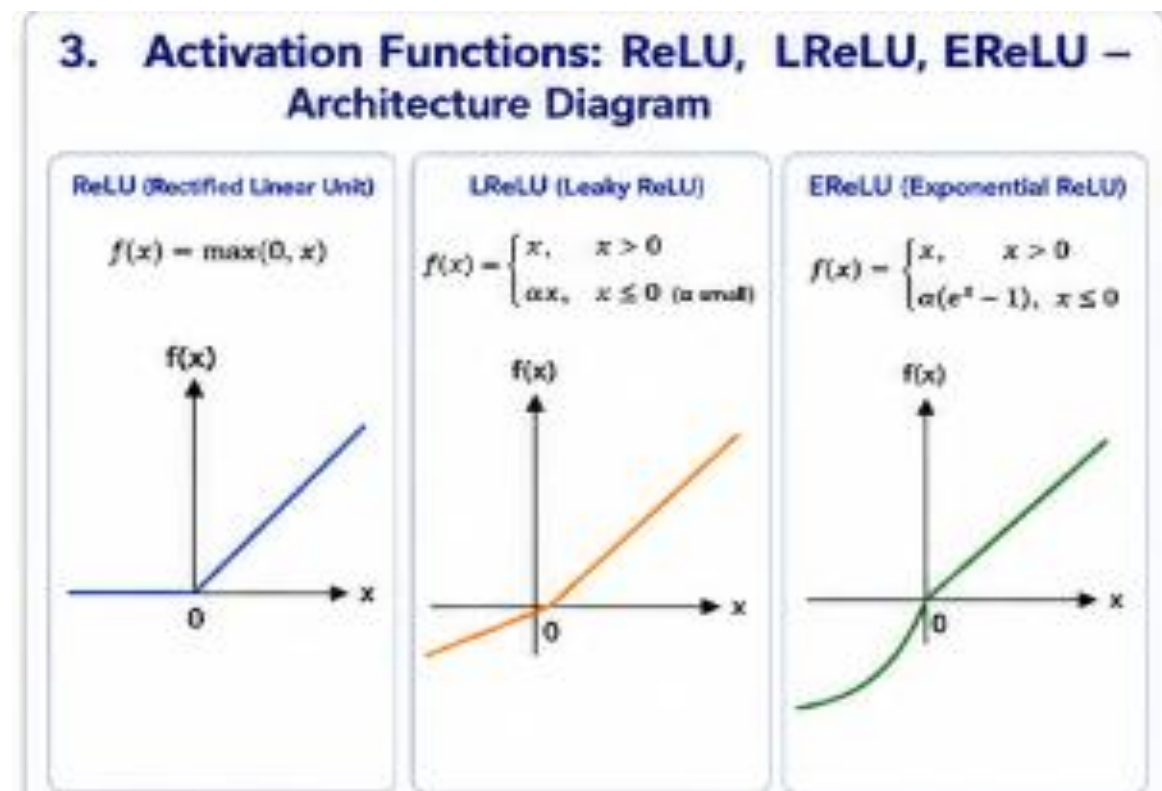
Activation functions introduce **non-linearity**, allowing neural networks to learn complex patterns.

Function	Basic Idea	Advantage	Limitation
ReLU	Negative $\rightarrow 0$, positive $\rightarrow x$	Simple and fast	Can cause dying neurons
LReLU	Negative values get a small slope	Reduces dying ReLU problem	Requires choosing slope
EReLU	Uses exponential behavior for negative values	Smooth negative region	Slightly more computation

ReLU $\rightarrow$ removes negative values

LReLU $\rightarrow$ keeps a small negative value

EReLU $\rightarrow$ smooth negative region



4. RBMs vs Autoencoders

Both can learn from unlabeled data, so they are examples of unsupervised learning.

RBM

A Restricted Boltzmann Machine contains:

- Visible layer
- Hidden layer

The visible layer receives the input, while the hidden layer learns hidden patterns in the data.

RBMs are probabilistic models and learn the probability distribution of the input data.

Applications include:

- Recommendation systems
- Feature learning
- Dimensionality reduction

Autoencoder

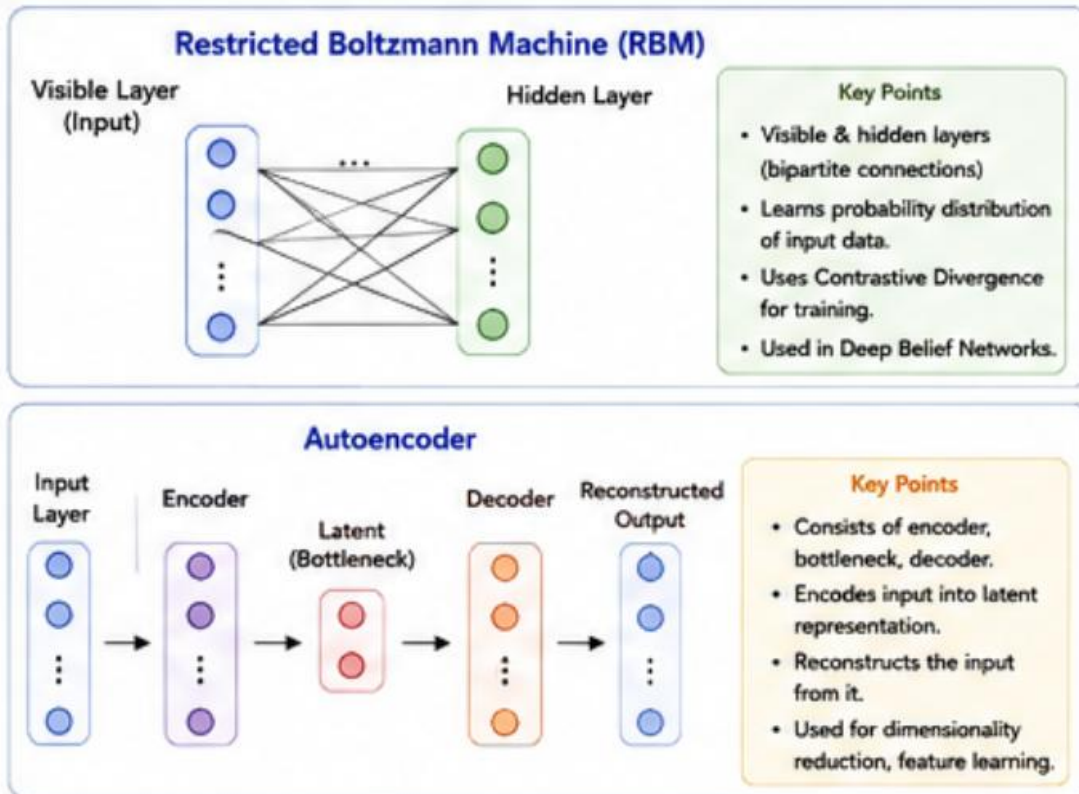
An autoencoder contains:

- Input layer
- Encoder
- Latent/bottleneck layer
- Decoder
- Reconstructed output

RBM → learns probability patterns

Autoencoder → compresses and reconstructs data

4. Unsupervised Training: RBMs vs Autoencoders – Architecture Diagram



5. Major Applications of Deep Learning

1. Computer Vision

Deep learning processes images and videos.

Applications:

- Image classification
- Object detection
- Face recognition
- Image segmentation

2. Natural Language Processing

Deep learning understands and generates human language.

Applications:

- Machine translation
- Chatbots
- Sentiment analysis

- Text summarization
- Question answering

3. Healthcare

Deep learning helps analyze medical information.

Applications:

- X-ray analysis
- MRI/CT analysis
- Disease detection
- Drug discovery
- Patient monitoring

4. Speech Recognition

Deep learning converts speech into meaningful information.

Applications:

- Speech-to-text
- Voice assistants
- Speaker identification
- Language recognition

5. Autonomous Systems

Deep learning allows machines to understand their environment and make decisions.

Applications:

- Self-driving cars
- Robot navigation
- Obstacle detection
- Path planning
- Autonomous drones

5. Applications of Deep Learning – Architecture Diagram

